

DISCRETE-TIME FILTER DESIGN

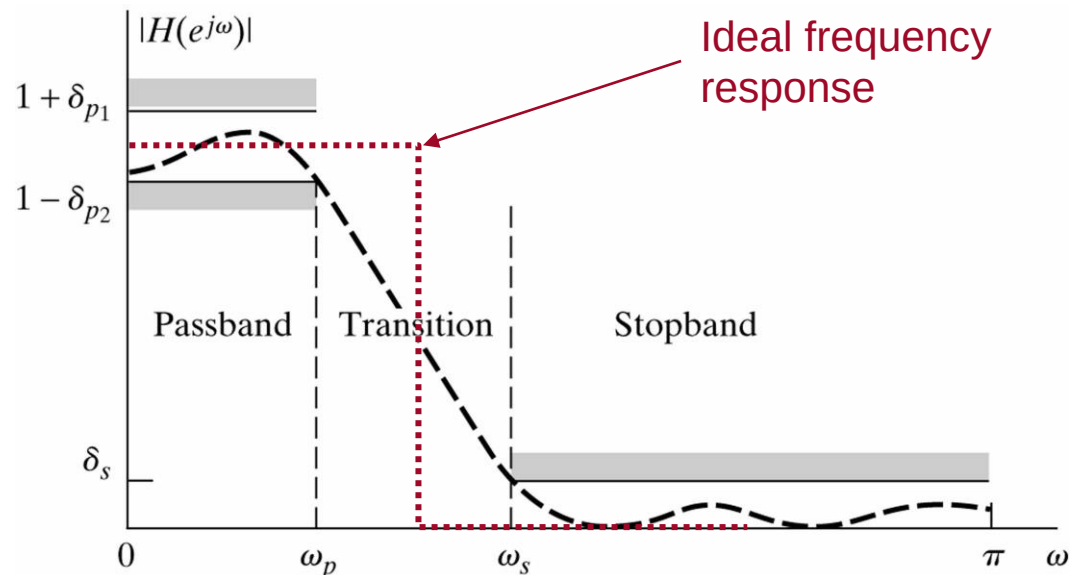
ENSC 429 – Digital Signal Processing

School of Engineering Science
Simon Fraser University
Burnaby, BC, Canada



DT FILTER DESIGN

- Filter design is a set of approaches for obtaining the coefficients of the filter's system function $H(z)$
- Ideal filters cannot be realized, so filter design is really about approximating the ideal frequency response using realizable structures
- Allowed deviations from ideal frequency response are specified in terms of tolerances



DT FILTER SPECIFICATION EXAMPLE

With sampling period $T = 10^{-4}$ seconds, design a DT filter so that the overall CT filter has

$$H_{\text{eff}}(j\Omega) = \begin{cases} 1 \pm 0.01 & |\Omega| \leq 4000\pi \text{ Hz} \\ \leq 0.001 & |\Omega| \geq 6000\pi \text{ Hz} \end{cases}$$

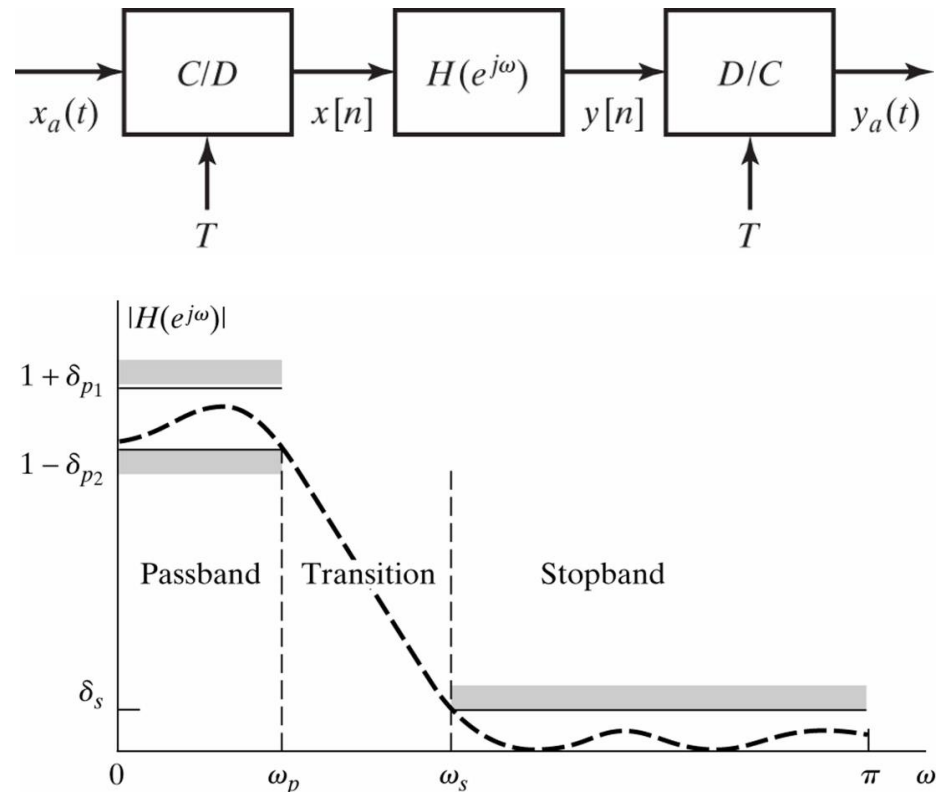
Mapping these to the specification symbols from the diagram:

$$\delta_{p1} = \delta_{p2} = 0.01$$

$$\delta_s = 0.001$$

$$\omega_p = 4000\pi \cdot 10^{-4} = 0.4\pi$$

$$\omega_s = 6000\pi \cdot 10^{-4} = 0.6\pi$$



Passband/stopband gains are often specified in terms of decibels (dB):

- Maximum passband gain: $20 \cdot \log_{10}(1 + 0.01) = 0.0864 \text{ dB}$
- Minimum passband gain: $20 \cdot \log_{10}(1 - 0.01) = -0.0873 \text{ dB}$
- Maximum stopband gain: $20 \cdot \log_{10}(0.001) = -60 \text{ dB}$

DT IIR FILTER DESIGN

- DT IIR filter design is largely based on CT IIR filter design
 - CT IIR filter design was well-developed at the time DT filter design was just starting
- Two basic design techniques in this category:
 1. Impulse invariance
 2. Bilinear transformation
- Each can be based on one of the popular CT filter classes (see Appendix B in the textbook):
 - Butterworth
 - Chebyshev
 - Elliptic

DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

1. Start with CT system function (assume first-order poles, for simplicity)

$$H_c(s) = \sum_{k=1}^N \frac{A_k}{s - s_k}$$

2. The corresponding CT impulse response is

$$h_c(t) = \begin{cases} \sum_{k=1}^N A_k e^{s_k t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

3. Obtain DT impulse response by sampling CT impulse response with sampling period T_d

$$h[n] = T_d \cdot h_c(nT_d) = T_d \sum_{k=1}^N A_k e^{s_k n T_d} u[n] = \sum_{k=1}^N T_d A_k e^{(s_k T_d)n} u[n]$$

4. DT system function is

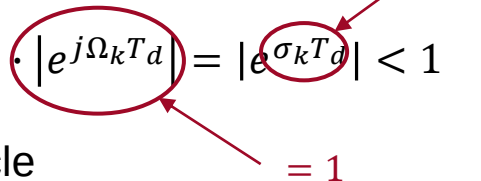
$$H(z) = \sum_{k=1}^N \frac{T_d A_k}{1 - e^{s_k T_d} z^{-1}}$$

DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

- The pole at $s = s_k$ in the s -plane became the pole at $z = e^{s_k T_d}$ in the z -plane
- If $s_k = \sigma_k + j\Omega_k$ was in the left half-plane ($\sigma_k < 0$), then

$$|z| = |e^{s_k T_d}| = |e^{\sigma_k T_d + j\Omega_k T_d}| = |e^{\sigma_k T_d}| \cdot |e^{j\Omega_k T_d}| = |e^{\sigma_k T_d}| < 1$$

so the pole in the z -plane is inside the unit circle



- Hence, a stable CT filter gives rise to a stable DT filter
- Also, partial fraction factors A_k of the CT system function became $T_d A_k$ in the DT system function
- Since impulse sampling can lead to potential aliasing, CT frequency response needs to be (sufficiently) bandlimited when using impulse invariance design

DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

- Example: design of a Butterworth DT filter by impulse invariance
- The general Butterworth CT system function is given by

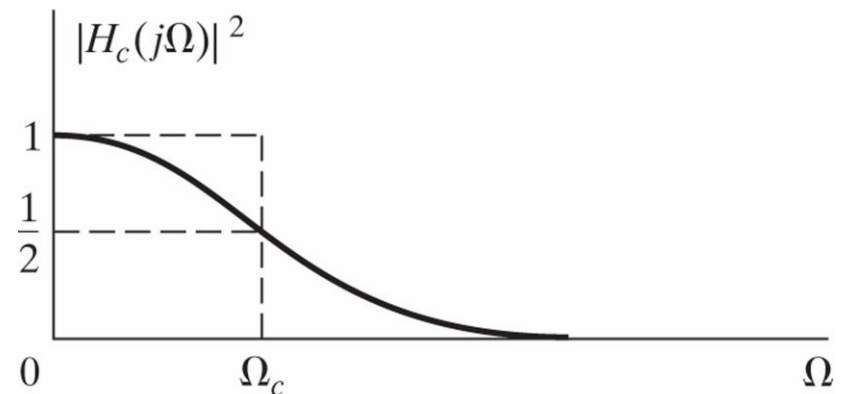
$$H_c(s) = \frac{c}{B_n(s)}$$

where c is a constant and $B_n(s)$ is a Butterworth polynomial of degree n

The magnitude-squared of a lowpass Butterworth filter is given by

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

where Ω_c is the “cutoff” frequency (where $|H_c(j\Omega_c)|^2 = 1/2$) and N is the filter order



DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

- Suppose our specifications in the DT frequency domain are

$$\begin{aligned} 0.89125 \leq |H(e^{j\omega})| \leq 1 & \quad 0 \leq |\omega| \leq 0.2\pi \\ |H(e^{j\omega})| \leq 0.17783 & \quad 0.3\pi \leq |\omega| \leq \pi \end{aligned}$$

- Note: $0.2\pi \leq |\omega| \leq 0.3\pi$ is the transition band, usually no specs are given there
- For simplicity, set $T_d = 1$ (the value of T_d actually does not matter) so $\omega = \Omega$
- This translates the DT specs to the following CT specs:

$$\begin{aligned} 0.89125 \leq |H_c(j\Omega)| \leq 1 & \quad 0 \leq |\Omega| \leq 0.2\pi \\ |H_c(j\Omega)| \leq 0.17783 & \quad 0.3\pi \leq |\Omega| \leq \pi \end{aligned}$$

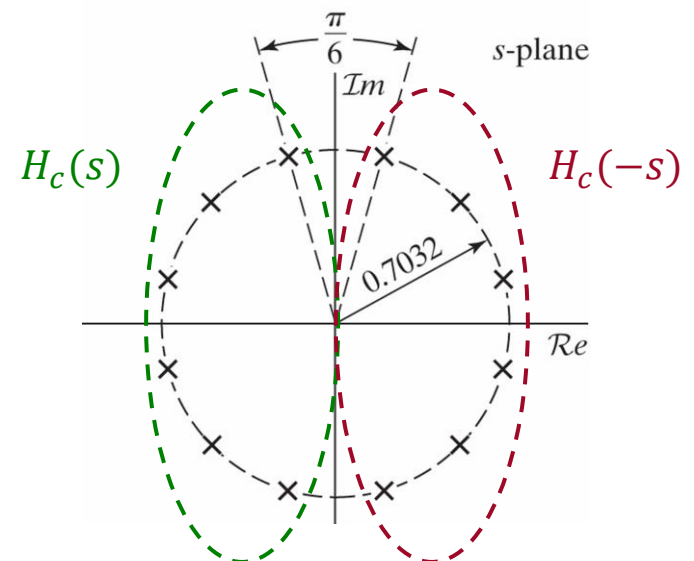
- Now have to solve

$$\begin{aligned} |H_c(j \cdot 0.2\pi)|^2 &= \frac{1}{1 + \left(\frac{0.2\pi}{\Omega_c}\right)^{2N}} = (0.89125)^2 \\ |H_c(j \cdot 0.3\pi)|^2 &= \frac{1}{1 + \left(\frac{0.3\pi}{\Omega_c}\right)^{2N}} = (0.17783)^2 \end{aligned}$$

DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

- Nonlinear system of two equations in two unknowns: Ω_c and N
- Solution (see notes): $N = 5.8858$, $\Omega_c = 0.70474$
 - Round N up to $N = 6$ (must be an integer), then recalculate from the first equation $\Omega_c = 0.7032$; this means the filter will meet passband specs and exceed stopband specs
 - If Ω_c is recalculated from the second equation, the filter would meet stopband specs and exceed passband specs
- With $N = 6$ and $\Omega_c = 0.7032$, we have

$$|H_c(s)|^2 = \frac{1}{1 + \left(\frac{s}{0.7032}\right)^{12}} = H_c(s)H_c(-s)$$
- 12 poles on a circle of radius 0.7032
- Poles of $H_c(s)$ are those in the left half-plane (stable filter)



DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

- With those six poles, $H_c(s)$ that satisfies $H_c(s)H_c(-s) = \frac{1}{1 + \left(\frac{s}{0.7032}\right)^{12}}$ is

$$H_c(s) = \frac{0.12093}{(s^2 + 0.3640s + 0.4945)(s^2 + 0.9945s + 0.4945)(s^2 + 1.3585s + 0.4945)}$$

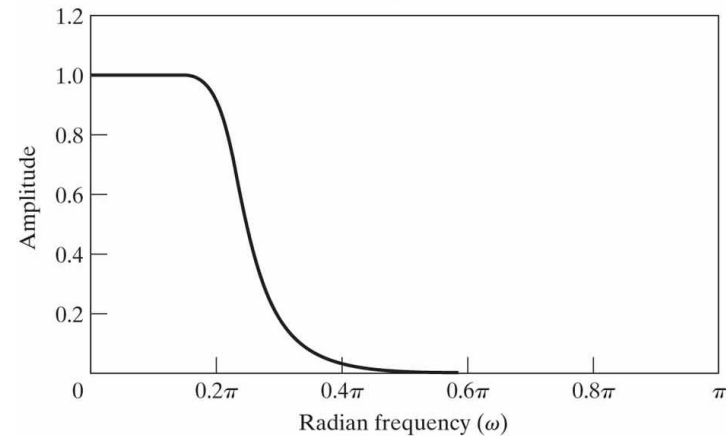
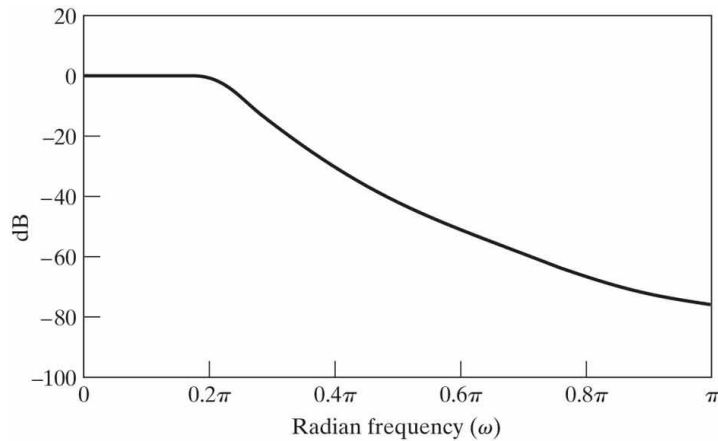
- After partial fraction expansion, following the procedure on slide 5, we get

$$H(z) = \frac{0.2871 - 0.4466z^{-1}}{1 - 1.2971z^{-1} + 0.6949z^{-2}} + \frac{-2.1428 - 1.1455z^{-1}}{1 - 1.0691z^{-1} + 0.3699z^{-2}} \\ + \frac{1.8557 - 0.6303z^{-1}}{1 - 0.9972z^{-1} + 0.2570z^{-2}}$$

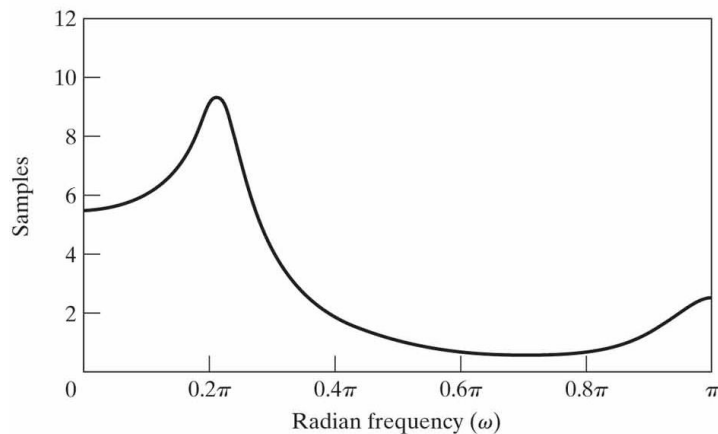
- Can be realized as a parallel connection of three second-order sections
- Other realizations are possible by rearranging $H(z)$

DT IIR FILTER DESIGN BY IMPULSE INVARIANCE

Magnitude response

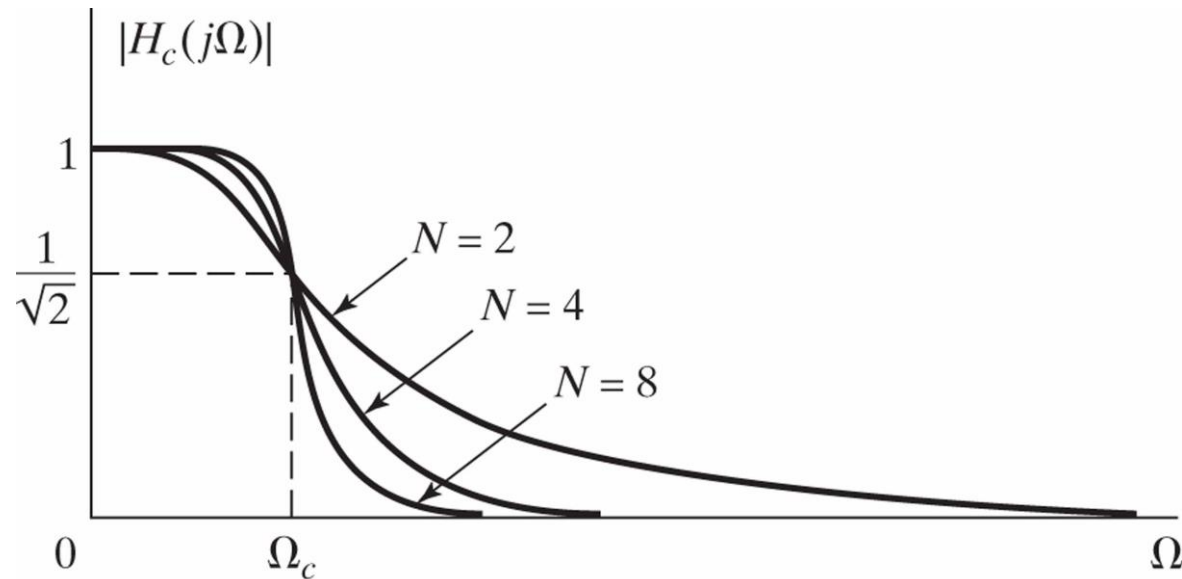


Group delay



BUTTERWORTH FILTER MAGNITUDE CHARACTERISTICS

Butterworth filters become more frequency-selective with increasing order N



But their implementation complexity also increases with N

DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- With impulse invariance design, aliasing of the CT frequency response could be an issue (when CT response is not sufficiently bandlimited)
- With bilinear transformation design, this is not an issue, as the entire CT frequency axis (from the s -plane) is mapped to one revolution of the unit circle (in the z -plane)

- The bilinear transformation between s and z is

$$s = \frac{2}{T_d} \cdot \frac{1 - z^{-1}}{1 + z^{-1}}$$

- Under this transformation, the DT system function is

$$H(z) = H_c \left(\frac{2}{T_d} \cdot \frac{1 - z^{-1}}{1 + z^{-1}} \right)$$

- Again, if the specs are given in the DT domain, the value of T_d does not matter, so it can be chosen for convenience

DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- Solving for z from the bilinear transformation and using $s = \sigma + j\Omega$

$$z = \frac{1 + s \frac{T_d}{2}}{1 - s \frac{T_d}{2}} = \frac{1 + \sigma \frac{T_d}{2} + j\Omega \frac{T_d}{2}}{1 - \sigma \frac{T_d}{2} - j\Omega \frac{T_d}{2}}$$

- Therefore ([see notes](#))

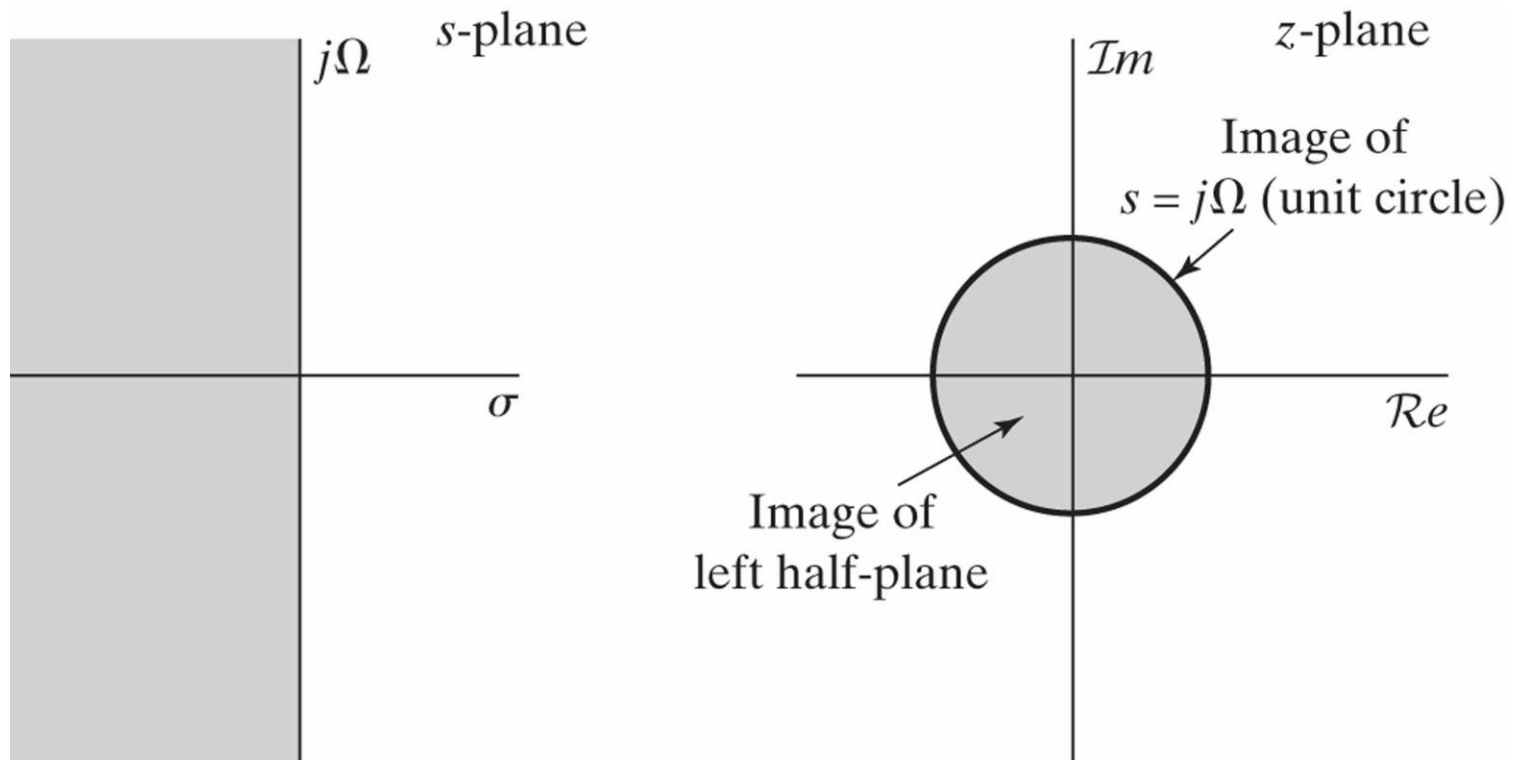
$$|z| = \begin{cases} > 1 & \sigma > 0 \\ = 1 & \sigma = 0 \\ < 1 & \sigma < 0 \end{cases}$$

- Left half s -plane maps to the interior of the unit circle in the z -plane
- Stable causal CT filters map to stable causal DT filters
- Relationship between CT and DT frequency ([see notes](#))

$$\omega = 2 \arctan\left(\Omega \frac{T_d}{2}\right)$$

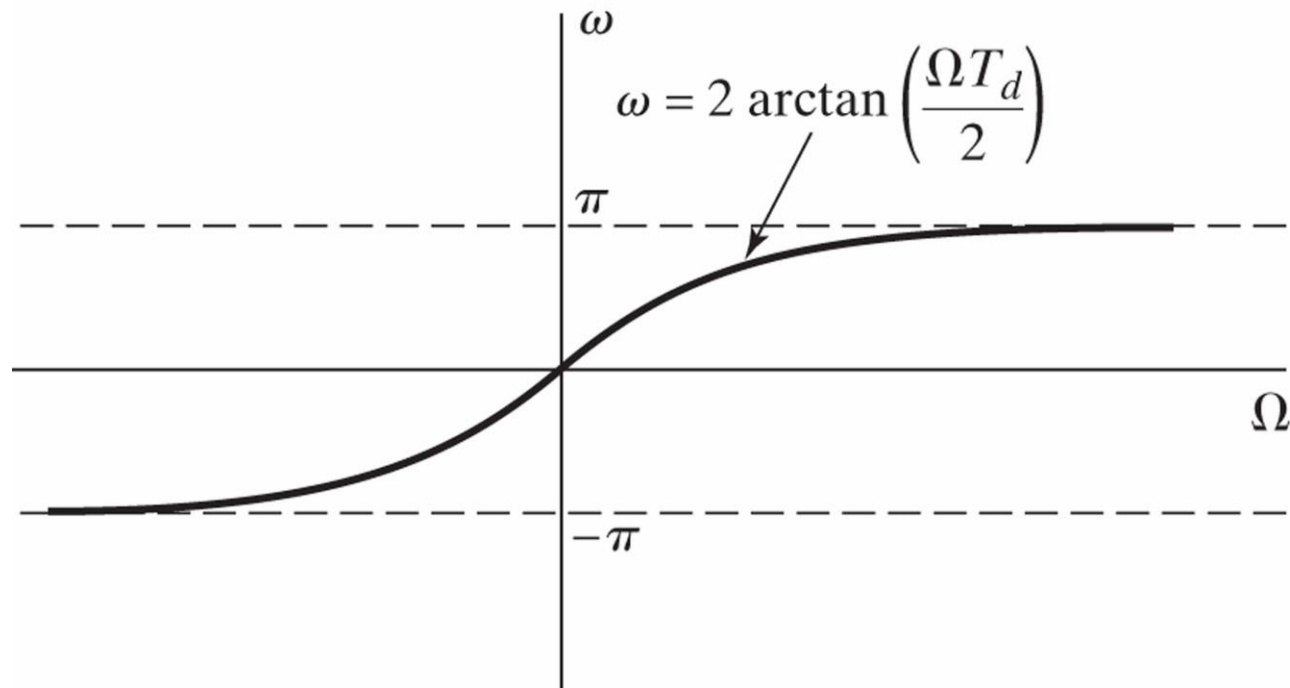
DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

s-plane to z-plane mapping



DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

CT to DT frequency mapping

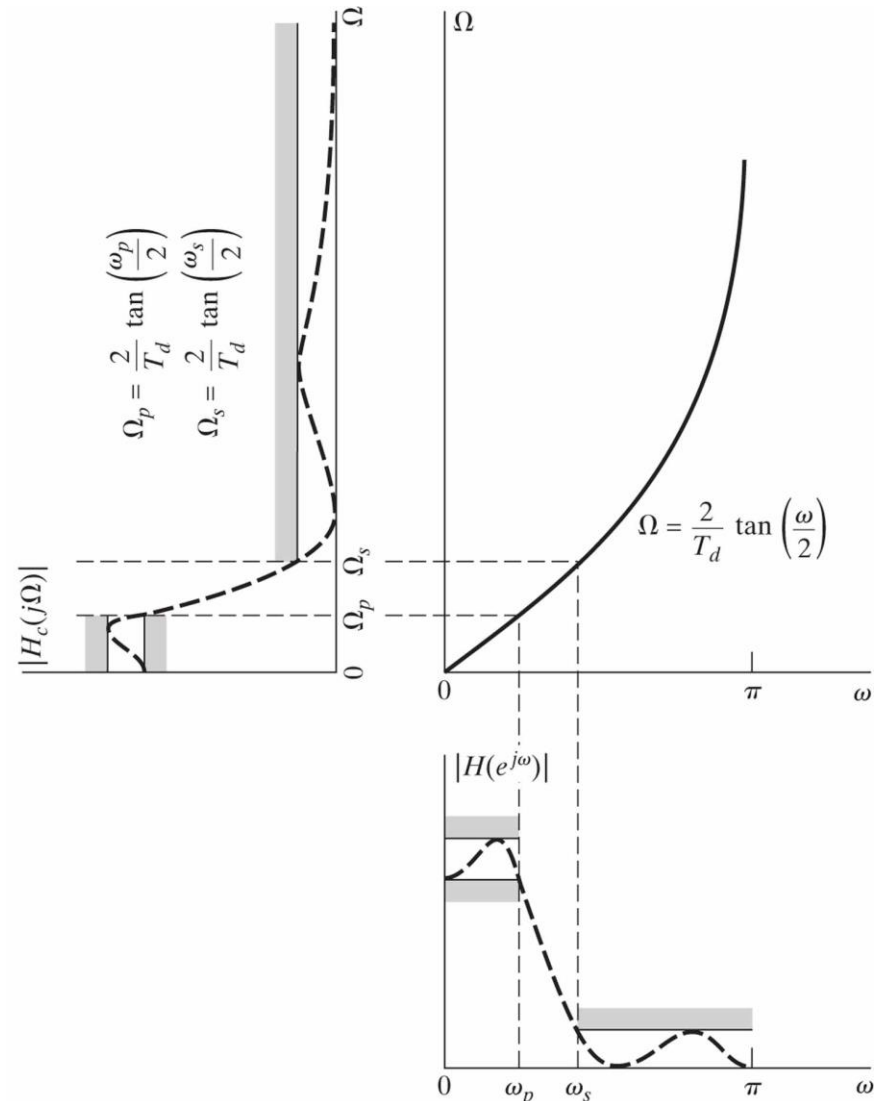


DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- The relationship between CT and DT frequency is nonlinear
- DT frequencies from the specs need to be pre-warped prior to CT filter design:

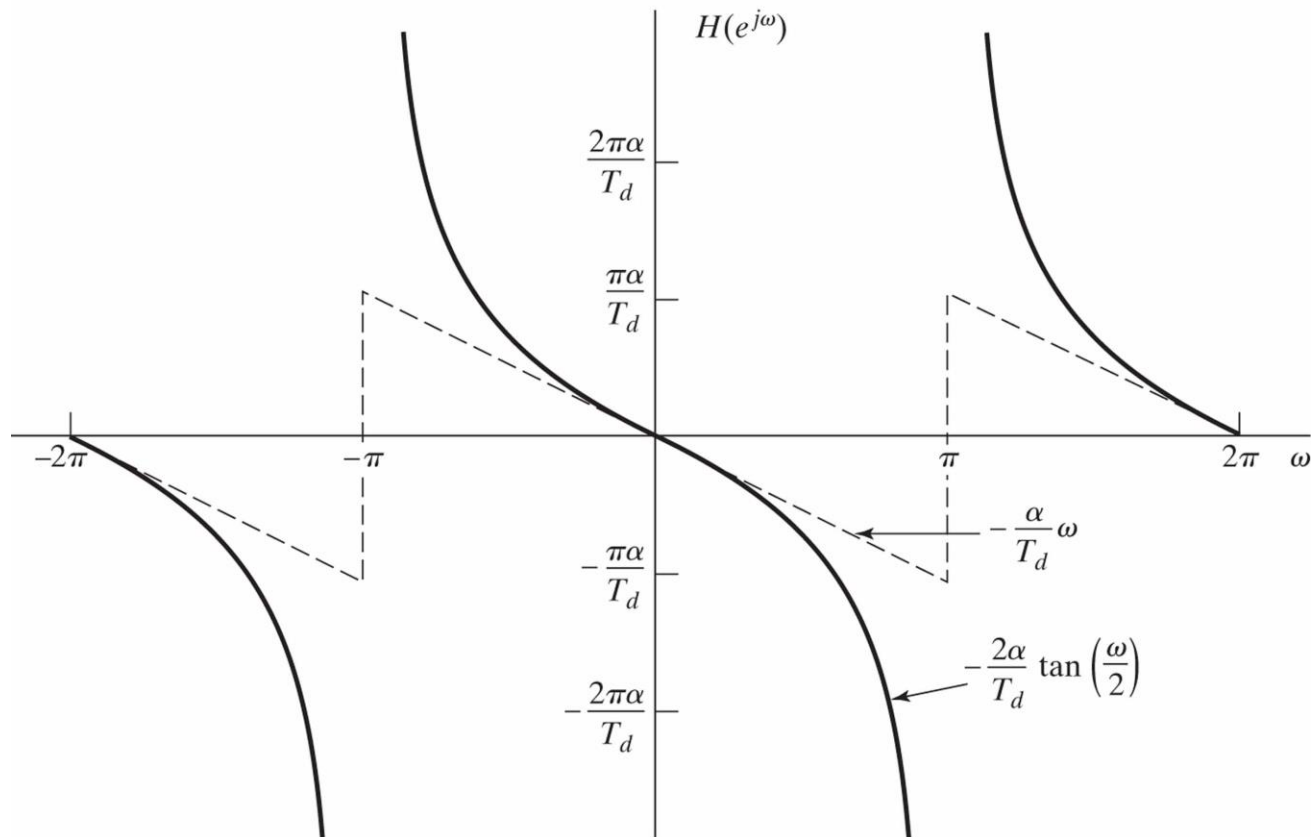
$$\Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

- The bilinear transformation will then warp CT axis back



DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

Due to nonlinear CT to DT frequency mapping, for DT phase response to be linear, CT phase response must be nonlinear



DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- Example (same as before): design of a Butterworth DT filter by bilinear transformation

$$\begin{aligned} 0.89125 \leq |H(e^{j\omega})| \leq 1 & \quad 0 \leq |\omega| \leq 0.2\pi \\ |H(e^{j\omega})| \leq 0.17783 & \quad 0.3\pi \leq |\omega| \leq \pi \end{aligned}$$

- First, need to pre-warp critical frequencies to find the corresponding CT frequencies:

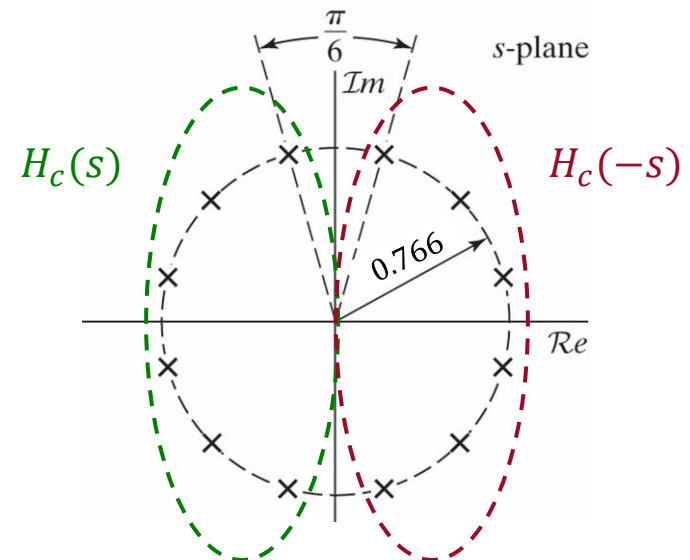
$$\begin{aligned} 0.89125 \leq |H_c(j\Omega)| \leq 1 & \quad 0 \leq |\Omega| \leq \frac{2}{T_d} \tan\left(\frac{0.2\pi}{2}\right) \\ |H_c(j\Omega)| \leq 0.17783 & \quad \frac{2}{T_d} \tan\left(\frac{0.3\pi}{2}\right) \leq |\Omega| \leq \infty \end{aligned}$$

- Setting $T_d = 1$ for convenience, we need to solve

$$\begin{aligned} |H_c(j \cdot 2 \tan(0.1\pi))|^2 &= \frac{1}{1 + \left(\frac{2 \tan(0.1\pi)}{\Omega_c}\right)^{2N}} = (0.89125)^2 \\ |H_c(j \cdot 2 \tan(0.15\pi))|^2 &= \frac{1}{1 + \left(\frac{2 \tan(0.15\pi)}{\Omega_c}\right)^{2N}} = (0.17783)^2 \end{aligned}$$

DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- Solution ([see notes](#)): $N = 5.305$
 - Round N up to $N = 6$ (must be an integer), then recalculate from the second equation this time $\Omega_c = 0.766$; this means the filter will meet stopband specs and exceed passband specs
 - If Ω_c is recalculated from the first equation, the filter would meet passband specs and exceed stopband specs
- With $N = 6$ and $\Omega_c = 0.766$, we have
$$|H_c(s)|^2 = \frac{1}{1 + \left(\frac{s}{0.766}\right)^{12}} = H_c(s)H_c(-s)$$
- 12 poles on a circle of radius 0.766
- Poles of $H_c(s)$ are those in the left half-plane (stable filter)



DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- With those six poles, $H_c(s)$ that satisfies $H_c(s)H_c(-s) = \frac{1}{1 + \left(\frac{s}{0.766}\right)^{12}}$ is

$$H_c(s) = \frac{0.20238}{(s^2 + 0.3996s + 0.5871)(s^2 + 1.0836s + 0.5871)(s^2 + 1.4802s + 0.5871)}$$

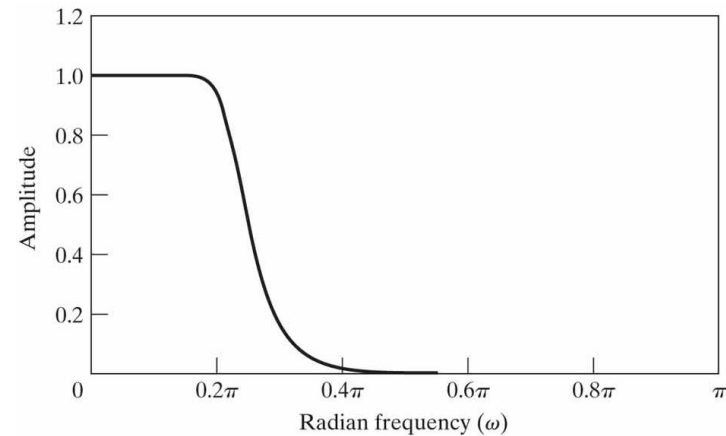
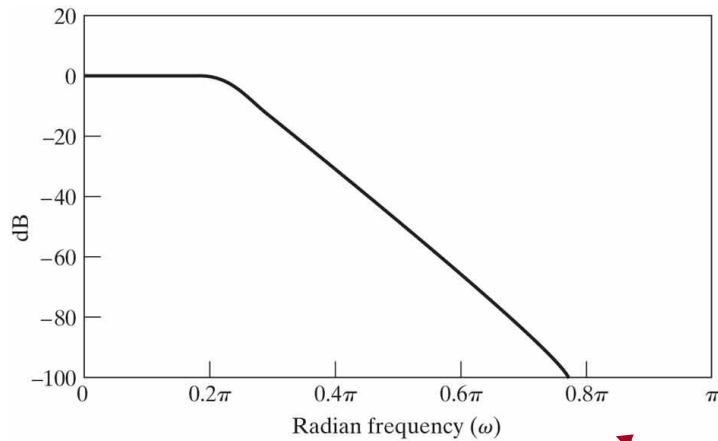
- After bilinear transformation $s = \frac{2}{T_d} \cdot \frac{1-z^{-1}}{1+z^{-1}}$, we get

$$H(z) = \frac{0.0007378(1 + z^{-1})^6}{(1 - 1.2686z^{-1} + 0.7051z^{-2})(1 - 1.0106z^{-1} + 0.3583z^{-2})} \cdot \frac{1}{1 - 0.9044z^{-1} + 0.2155z^{-2}}$$

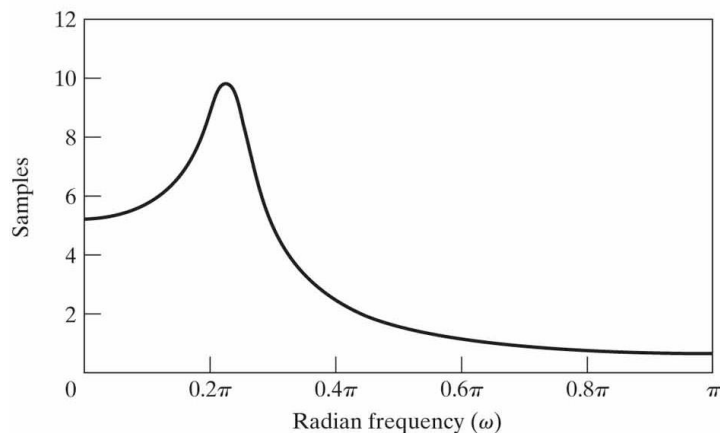
- Can be realized as a series connection of three second-order sections
- Other realizations are possible by rearranging $H(z)$

DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

Magnitude response



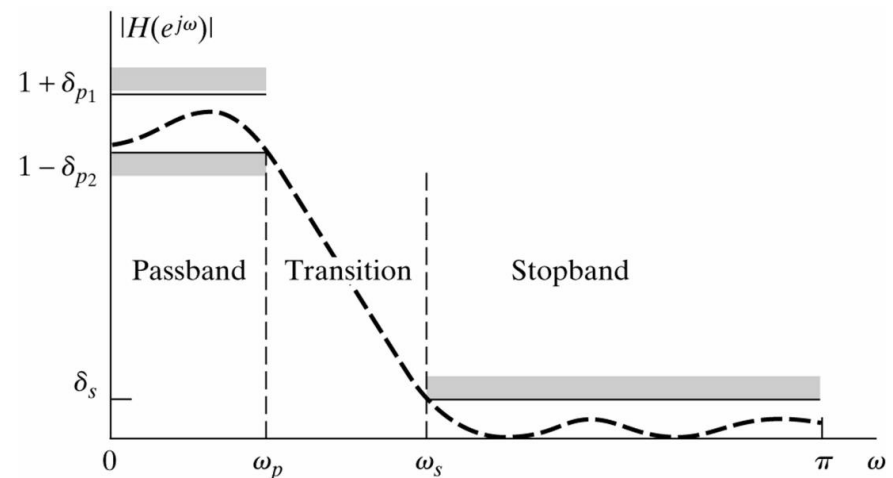
Group delay



Due to frequency compression caused by bilinear transformation, the magnitude response of this filter drops off much more rapidly than that of the filter designed by impulse invariance

DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

Example: comparison of Butterworth, Chebyshev, and elliptic filter designs by bilinear transformation

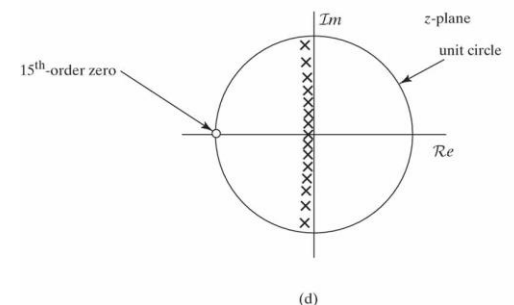
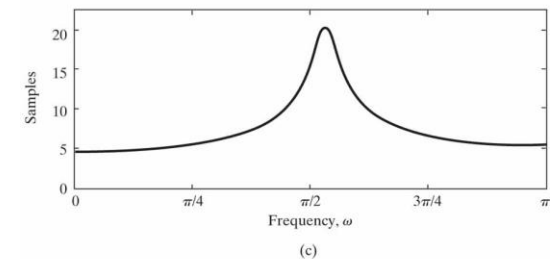
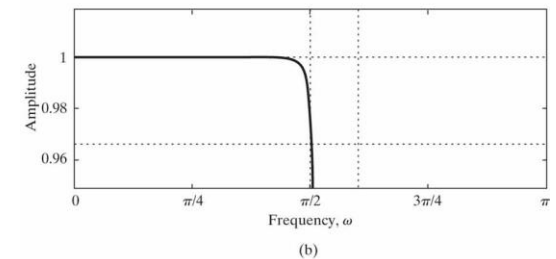
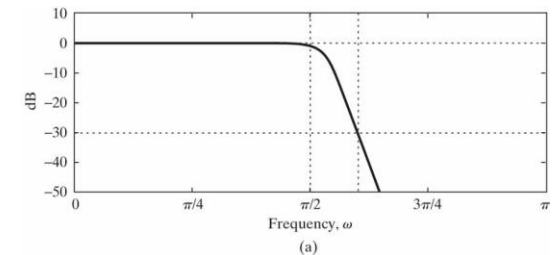


Specifications:

- Lowpass filter
- Passband edge frequency: $\omega_p = 0.5\pi$
- Stopband edge frequency: $\omega_s = 0.6\pi$
- Maximum passband gain: $20 \cdot \log_{10}(1 + \delta_{p1}) = 0 \text{ dB} \Rightarrow \delta_{p1} = 0$
- Minimum passband gain: $20 \cdot \log_{10}(1 - \delta_{p2}) = -0.3 \text{ dB} \Rightarrow \delta_{p2} = 0.0339$
- Maximum stopband gain: $20 \cdot \log_{10}(\delta_s) = -30 \text{ dB} \Rightarrow \delta_s = 0.0316$

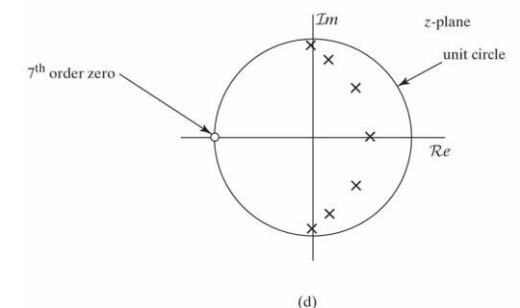
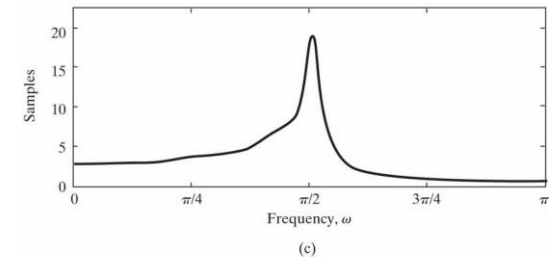
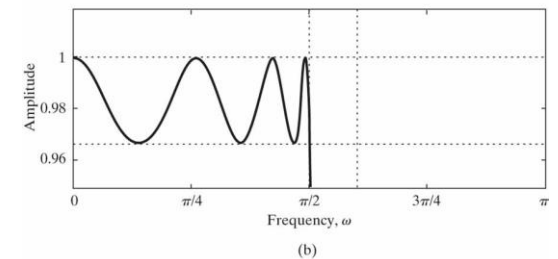
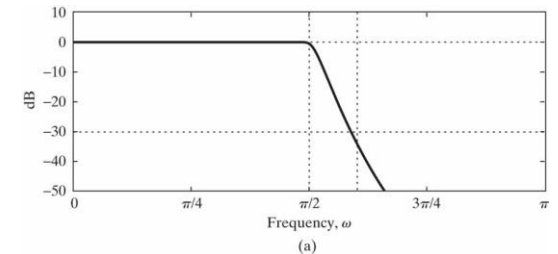
DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- Use Matlab! (industry standard for these types of problems)
- Check the following functions: `butter`, `cheby1`, `cheby2`, `ellip`, `freqz`, `fdesign`
- For the Butterworth filter, minimum order to meet the specs is $N = 15$



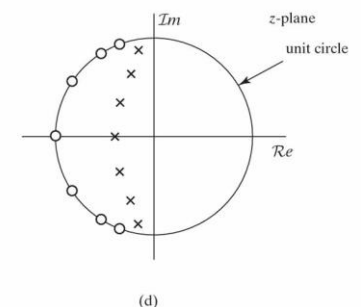
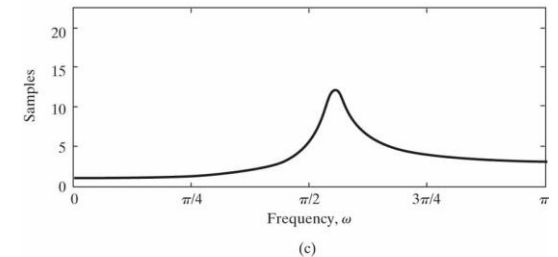
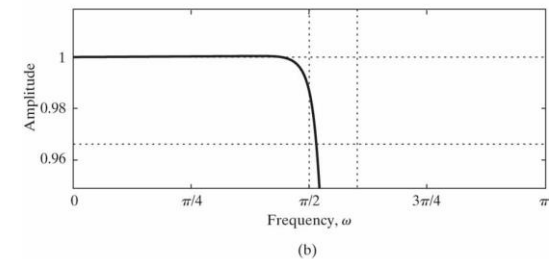
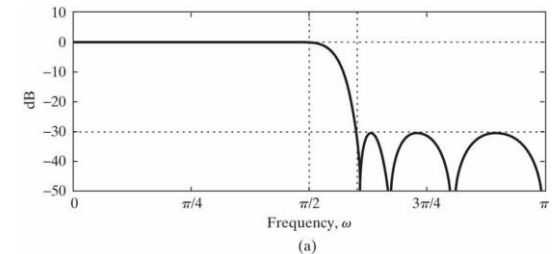
DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- For the Chebyshev type I filter, minimum order to meet the specs is $N = 7$
- Allowing passband ripple allows for a lower order filter compared to Butterworth



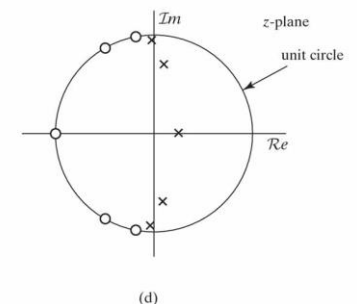
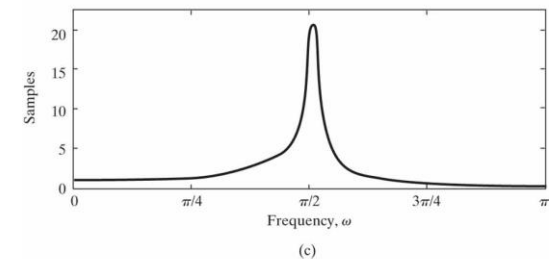
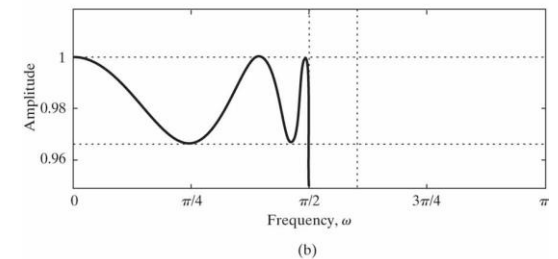
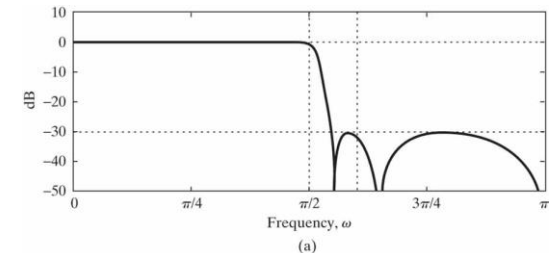
DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

- For the Chebyshev type II filter, minimum order to meet the specs is again $N = 7$
- Allowing stopband ripple allows for a lower order filter compared to Butterworth



DT IIR FILTER DESIGN BY BILINEAR TRANSFORMATION

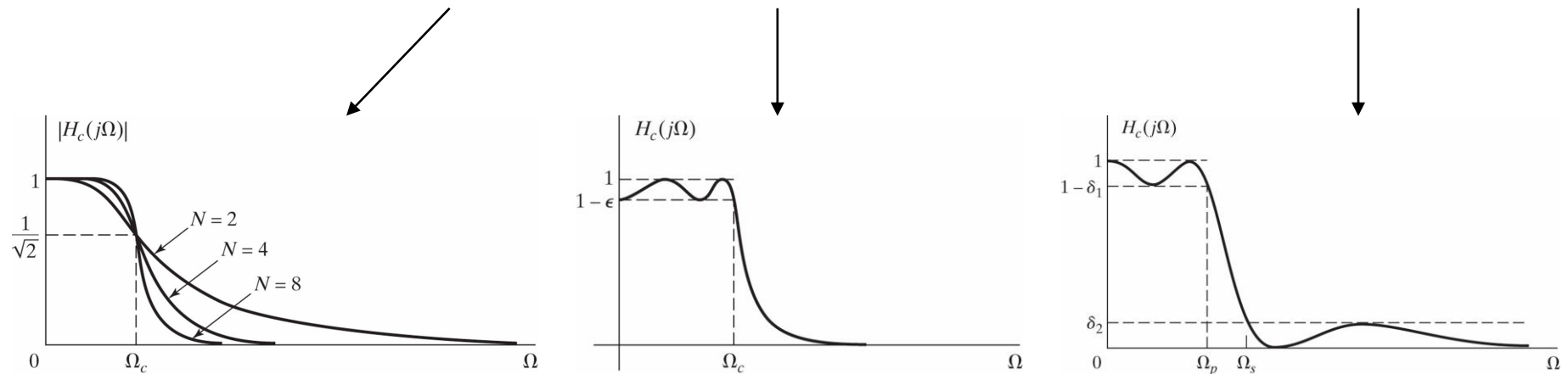
- For the elliptic filter, minimum order to meet the specs is again $N = 5$
- Allowing both passband and stopband ripple allows for a lower order filter even compared to Chebyshev filters



DT IIR FILTER DESIGN

Filter magnitude response characteristics

Filter	Butterworth	Chebyshev I	Chebyshev II	Elliptic
Passband	Monotonic	Equiripple	Monotonic	Equiripple
Stopband	Monotonic	Monotonic	Equiripple	Equiripple
Order (for the same specs)	Highest	Medium	Medium	Lowest

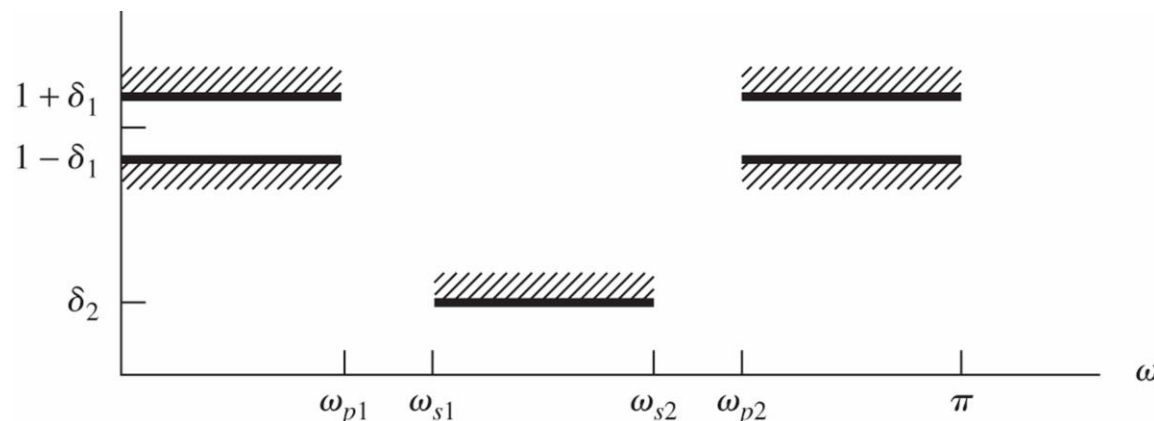


DT FILTER DESIGN BY FREQUENCY TRANSFORMATION

- So far, we only talked about lowpass filter design
- Other types of filters (highpass, bandpass, ...) can be obtained from a prototype lowpass filter $H_{lp}(Z)$ by frequency transformation
- This frequency transformation is of the form $Z^{-1} = G(z^{-1})$, that is

$$H(z) = H_{lp}(Z) \Big|_{Z^{-1}=G(z^{-1})}$$

- Generally, $G(z^{-1})$ is chosen to preserve passband and stopband behavior (tolerances) and change the critical frequencies



DT FILTER DESIGN BY FREQUENCY TRANSFORMATION

Table 7.1 – transformations from a lowpass filter with passband cutoff frequency θ_p to another filter with desired passband cutoff frequencies:

- ω_p for high-pass filters
- ω_{p1} and ω_{p2} for band-pass and band-stop filters

Filter Type	Transformations	Associated Design Formulas
Lowpass	$Z^{-1} = \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$	$\alpha = \frac{\sin\left(\frac{\theta_p - \omega_p}{2}\right)}{\sin\left(\frac{\theta_p + \omega_p}{2}\right)}$ $\omega_p = \text{desired cutoff frequency}$
Highpass	$Z^{-1} = -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$	$\alpha = -\frac{\cos\left(\frac{\theta_p + \omega_p}{2}\right)}{\cos\left(\frac{\theta_p - \omega_p}{2}\right)}$ $\omega_p = \text{desired cutoff frequency}$
Bandpass	$Z^{-1} = -\frac{z^{-2} - \frac{2\alpha k}{k+1}z^{-1} + \frac{k-1}{k+1}}{\frac{k-1}{k+1}z^{-2} - \frac{2\alpha k}{k+1}z^{-1} + 1}$	$\alpha = \frac{\cos\left(\frac{\omega_{p2} + \omega_{p1}}{2}\right)}{\cos\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right)}$ $k = \cot\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right) \tan\left(\frac{\theta_p}{2}\right)$ $\omega_{p1} = \text{desired lower cutoff frequency}$ $\omega_{p2} = \text{desired upper cutoff frequency}$
Bandstop	$Z^{-1} = \frac{z^{-2} - \frac{2\alpha}{1+k}z^{-1} + \frac{1-k}{1+k}}{\frac{1-k}{1+k}z^{-2} - \frac{2\alpha}{1+k}z^{-1} + 1}$	$\alpha = \frac{\cos\left(\frac{\omega_{p2} + \omega_{p1}}{2}\right)}{\cos\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right)}$ $k = \tan\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right) \tan\left(\frac{\theta_p}{2}\right)$ $\omega_{p1} = \text{desired lower cutoff frequency}$ $\omega_{p2} = \text{desired upper cutoff frequency}$

DT FILTER DESIGN BY FREQUENCY TRANSFORMATION

- Example: transforming a lowpass filter to a high-pass filter
- A lowpass filter with passband cutoff frequency $\theta_p = 0.2\pi$ has the system function

$$H(Z) = \frac{0.001836(1 + Z^{-1})^4}{(1 - 1.5548Z^{-1} + 0.6493Z^{-2})(1 - 1.4996Z^{-1} + 0.8482Z^{-2})}$$

- We want to transform it to a high-pass filter with cutoff frequency $\omega_p = 0.6\pi$
- From the table

$$\alpha = \frac{\cos\left(\frac{\theta_p + \omega_p}{2}\right)}{\cos\left(\frac{\theta_p - \omega_p}{2}\right)} = \frac{\cos\left(\frac{0.2\pi + 0.6\pi}{2}\right)}{\cos\left(\frac{0.2\pi - 0.6\pi}{2}\right)} = -0.38197$$

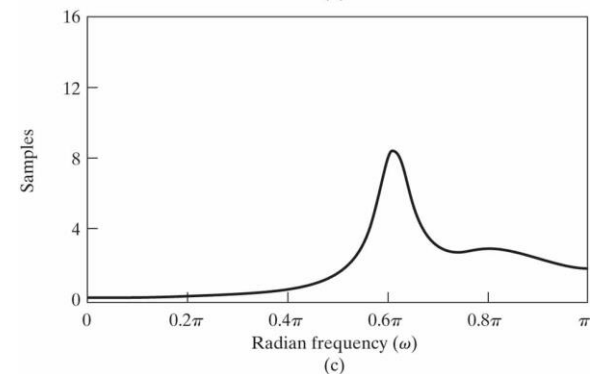
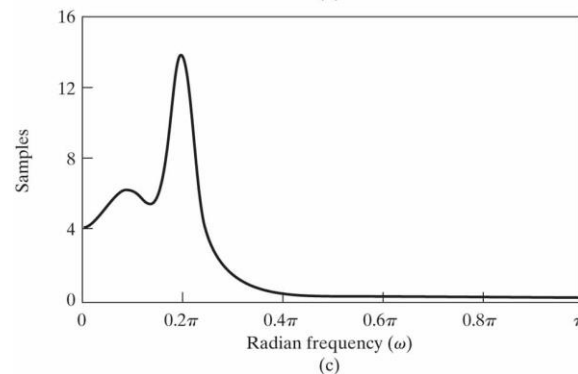
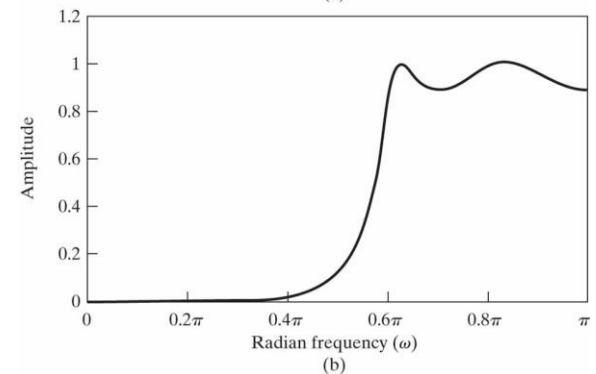
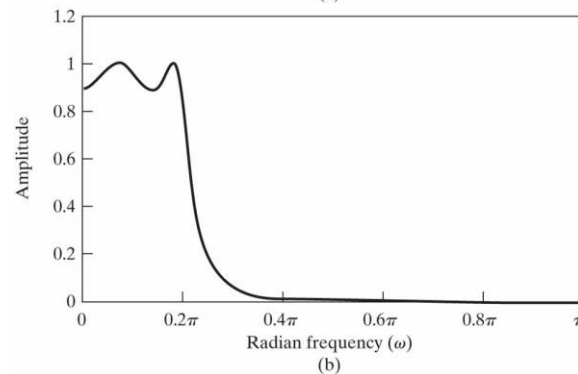
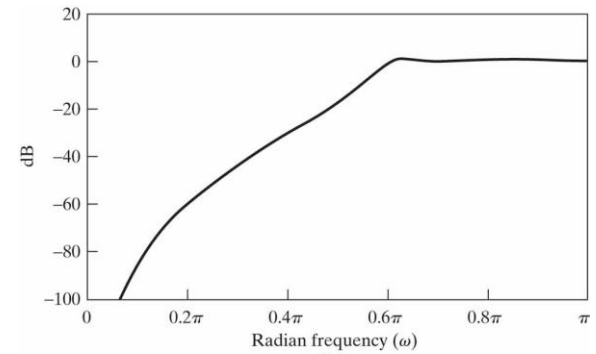
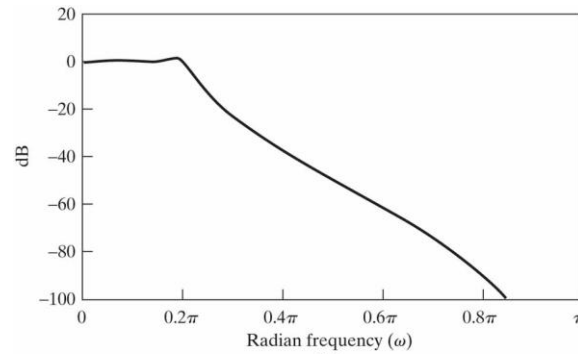
- Using this, we get

$$\begin{aligned} H(z) &= H_{lp}(Z) \Big|_{Z^{-1} = -[(z^{-1} - 0.38197)/(1 - 0.38197z^{-1})]} \\ &= \frac{0.02426(1 + z^{-1})^4}{(1 + 1.0416z^{-1} + 0.4019z^{-2})(1 + 0.5661z^{-1} + 0.7657z^{-2})} \end{aligned}$$

DT FILTER DESIGN BY FREQUENCY TRANSFORMATION

Left: lowpass

Right: high-pass



DT FIR FILTER DESIGN

- Popular methods for FIR filter design are:
 - Windowing
 - Direct optimization
- Design by windowing is conceptually simpler
- It involves the following steps
 1. Start with a “desired” system function $H_d(z)$ or frequency response $H_d(e^{j\omega})$
 2. Find the “desired” impulse response $h_d[n]$
 3. Apply a window function $w[n]$ to obtain

$$h[n] = h_d[n] \cdot w[n]$$

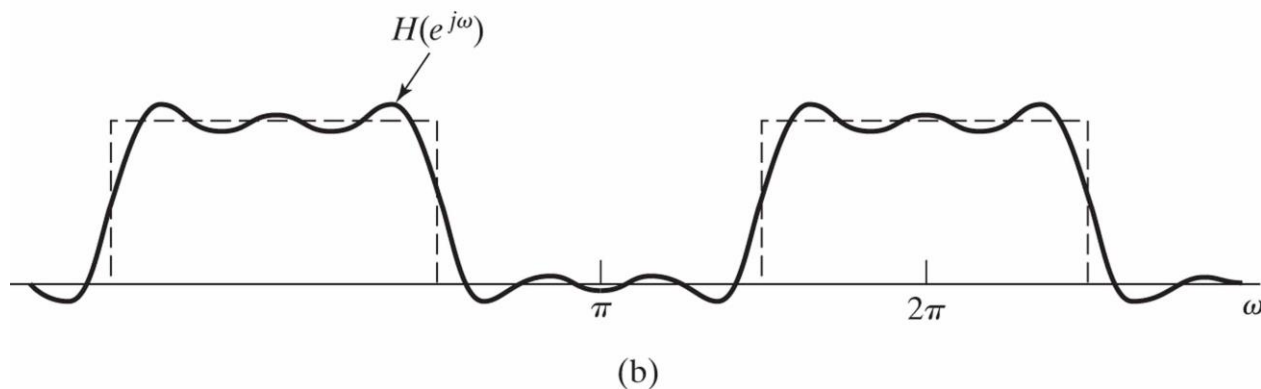
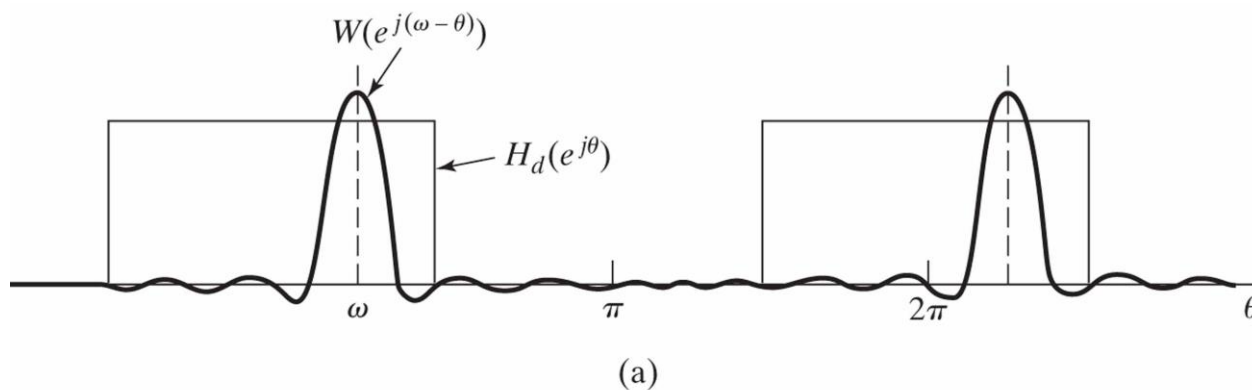
- Recall: multiplication in time domain is convolution in frequency domain:

$$H(e^{j\omega}) = H_d(e^{j\omega}) \circledast W(e^{j\omega})$$

- So, the final frequency response $H(e^{j\omega})$ is a “smeared” version of $H_d(e^{j\omega})$

DT FIR FILTER DESIGN BY WINDOWING

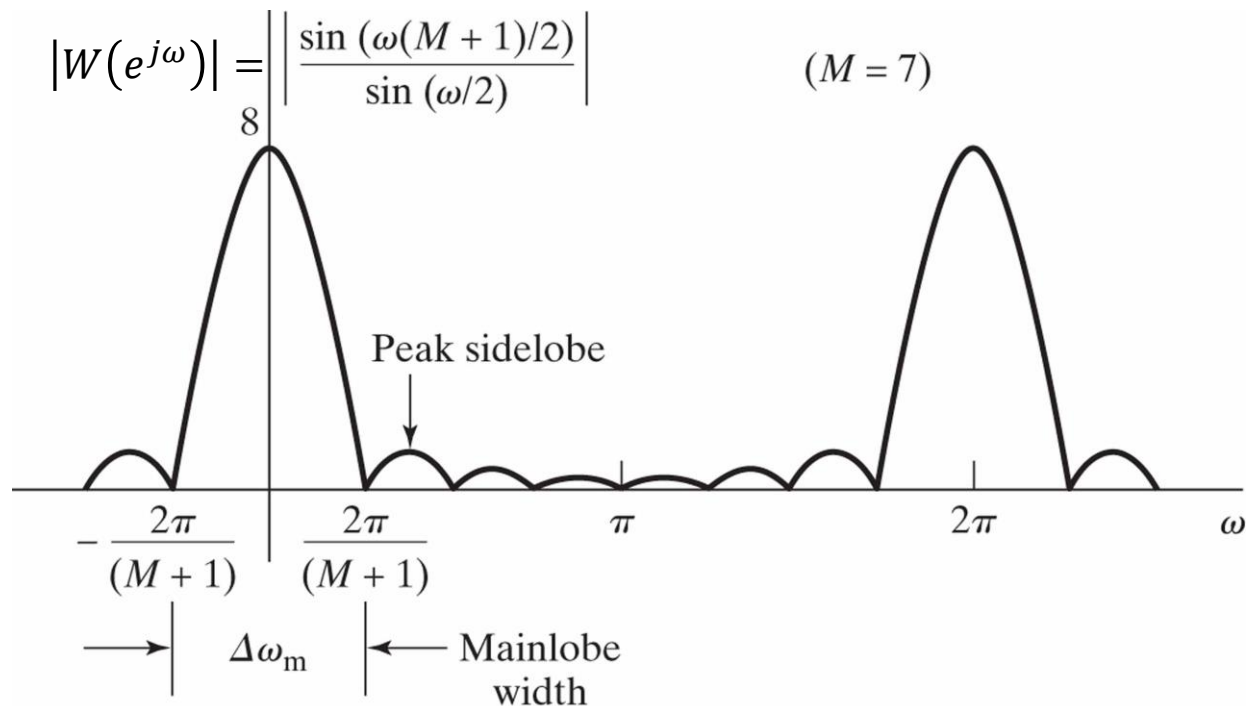
Effect of windowing in the frequency domain



DT FIR FILTER DESIGN BY WINDOWING

The simplest window is the **rectangular** window

$$w[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$



DT FIR FILTER DESIGN BY WINDOWING

Other popular windows:

- Bartlett (triangular) for even M

$$w[n] = \begin{cases} 2n/M & 0 \leq n \leq M/2 \\ 2 - 2n/M & M/2 < n \leq M \\ 0 & \text{else} \end{cases}$$

- Hann

$$w[n] = \begin{cases} 0.5 - 0.5 \cos(2\pi n/M) & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$

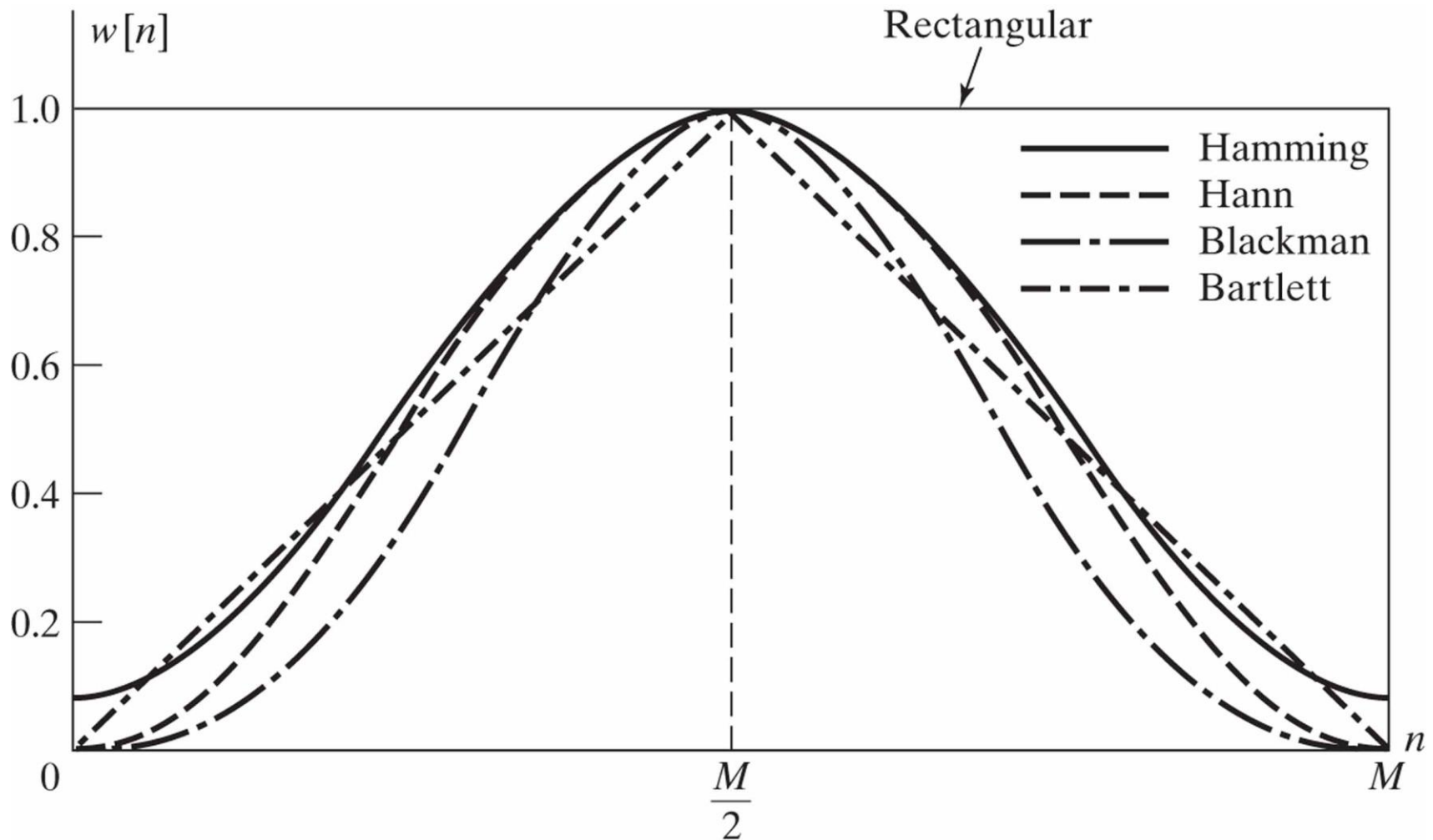
- Hamming

$$w[n] = \begin{cases} 0.54 - 0.46 \cos(2\pi n/M) & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$

- Blackman

$$w[n] = \begin{cases} 0.42 - 0.5 \cos(2\pi n/M) + 0.08 \cos(4\pi n/M) & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$

DT FIR FILTER DESIGN BY WINDOWING

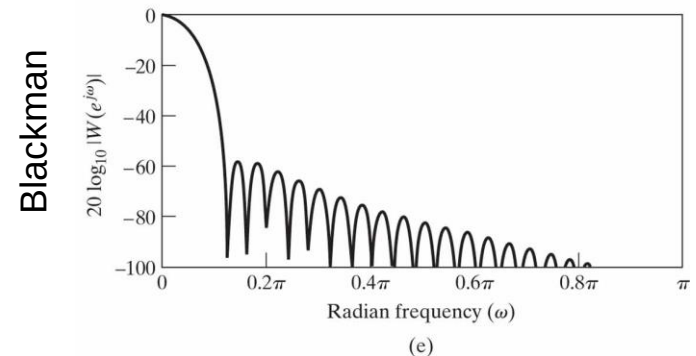
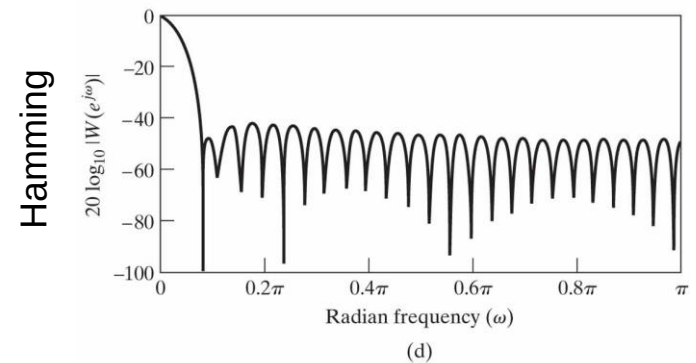
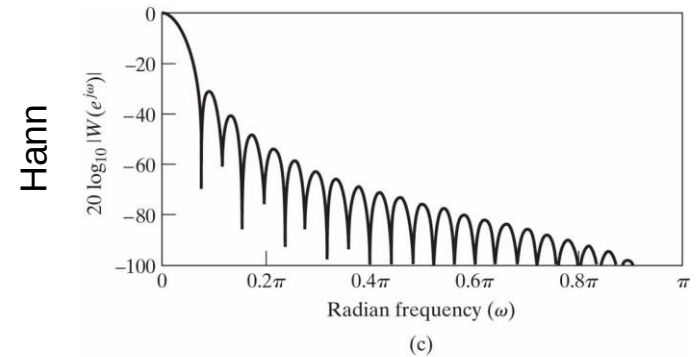
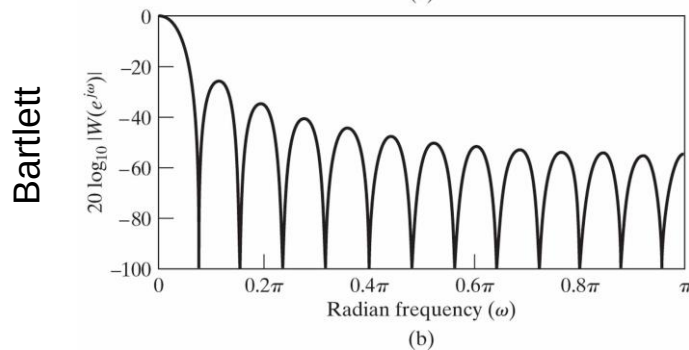
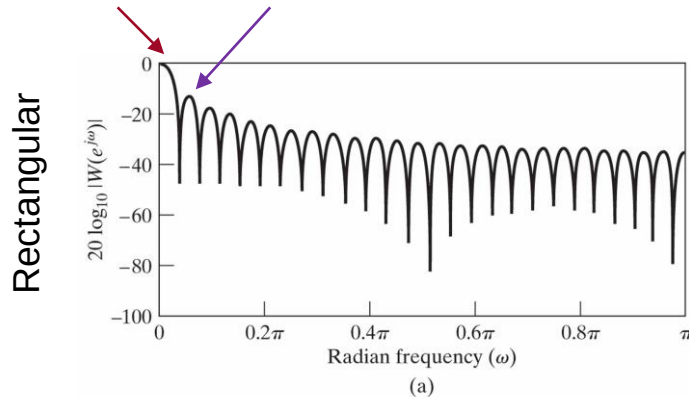


DT FIR FILTER DESIGN BY WINDOWING

Magnitude responses of various windows
for $M = 50$

Main lobe
(want narrow)

Side lobe
(want suppressed)



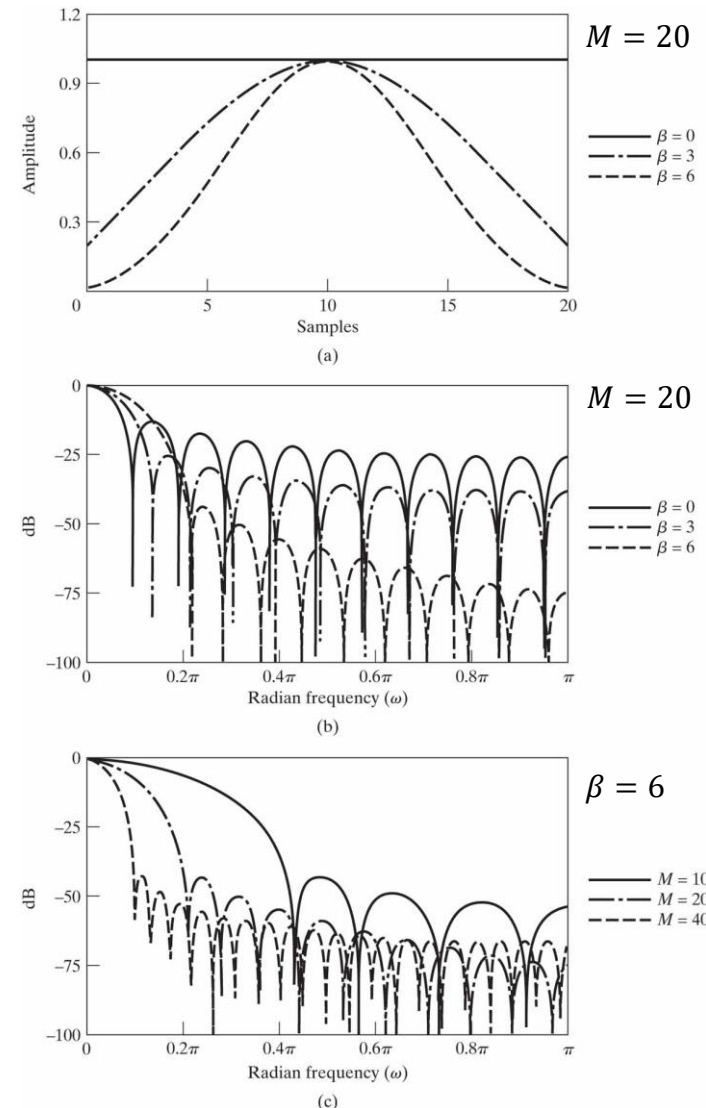
DT FIR FILTER DESIGN BY WINDOWING

- **Kaiser window** is defined as

$$w[n] = \begin{cases} \frac{I_0 \left[\beta \sqrt{1 - \left[\frac{n - \alpha}{\alpha} \right]^2} \right]}{I_0(\beta)} & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$

where $\alpha = M/2$ and I_0 is the zeroth-order modified Bessel function of the first kind

- Changing the parameters M (length) and β (shape) allows exploring the trade-off between main lobe width and side lobe height



DT FIR FILTER DESIGN BY WINDOWING

Kaiser window design (for a lowpass filter)

- Let ω_p and ω_s be passband and stopband critical frequencies, respectively, and δ be the passband and stopband tolerance

- Define

$$\Delta\omega = \omega_s - \omega_p \quad \text{and} \quad A = -20 \cdot \log_{10} \delta$$

- Then

$$\beta = \begin{cases} 0.1102(A - 8.7) & A > 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 \leq A \leq 50 \\ 0 & A < 21 \end{cases}$$

and

$$M = \frac{A - 8}{2.285 \cdot \Delta\omega}$$

- Kaiser window with these values of β and M , when applied to an ideal lowpass filter impulse response, will meet the specifications

DT FIR FILTER DESIGN BY WINDOWING

Example: lowpass filter design using the Kaiser window approach

- Let $\omega_p = 0.4\pi$, $\omega_s = 0.6\pi$, $\delta_{p1} = \delta_{p2} = 0.01$, and $\delta_s = 0.001$.
- Since Kaiser window method requires one tolerance δ , we set $\delta = \min\{\delta_{p1}, \delta_{p2}, \delta_s\} = 0.001$, so that stopband tolerance is met and passband tolerance specs are exceeded
- As the desired lowpass filter to be approximated, we choose ideal lowpass filter with cutoff frequency $\omega_c = (\omega_p + \omega_s)/2$
- This filter has impulse response $h_d[n] = \frac{\sin(\omega_c n)}{\pi n}$. Since the window covers $0 \leq n \leq M$, we shift $h_d[n]$ by $\alpha = M/2$ to center it in that range: $h_d[n] = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)}$
- Using the equations from the previous slide, we find $\beta = 5.653$ and $M = 37$, so

$$h[n] = h_d[n] \cdot w[n] = \begin{cases} \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)} \cdot \frac{I_0\left[\beta \sqrt{\left(1 - \left[\frac{n-\alpha}{\alpha}\right]^2}\right)}\right]}{I_0(\beta)} & 0 \leq n \leq M \\ 0 & \text{else} \end{cases}$$

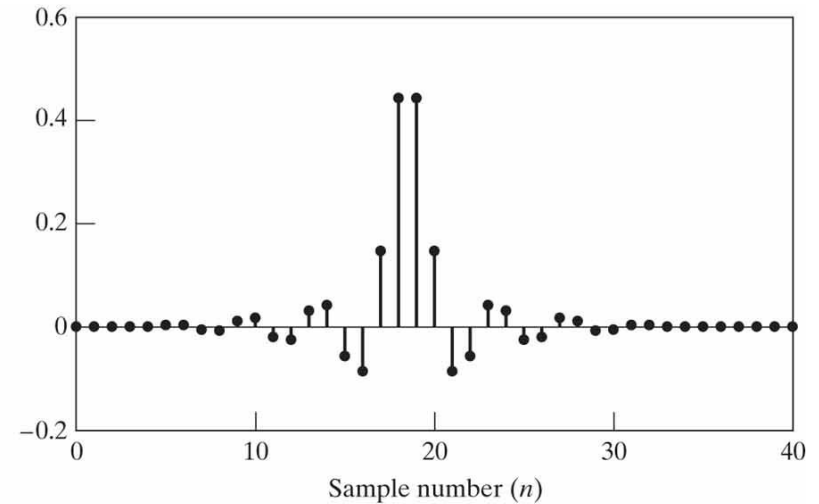
DT FIR FILTER DESIGN BY WINDOWING

Impulse response

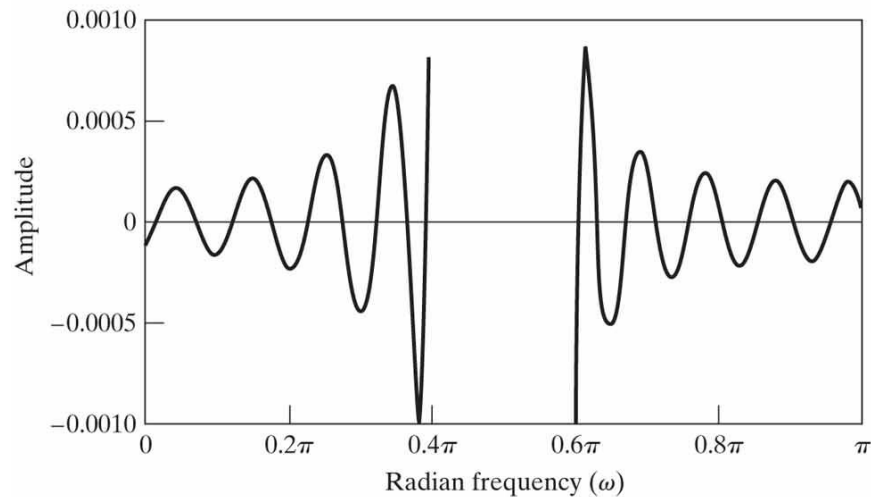
Magnitude response

Ripple

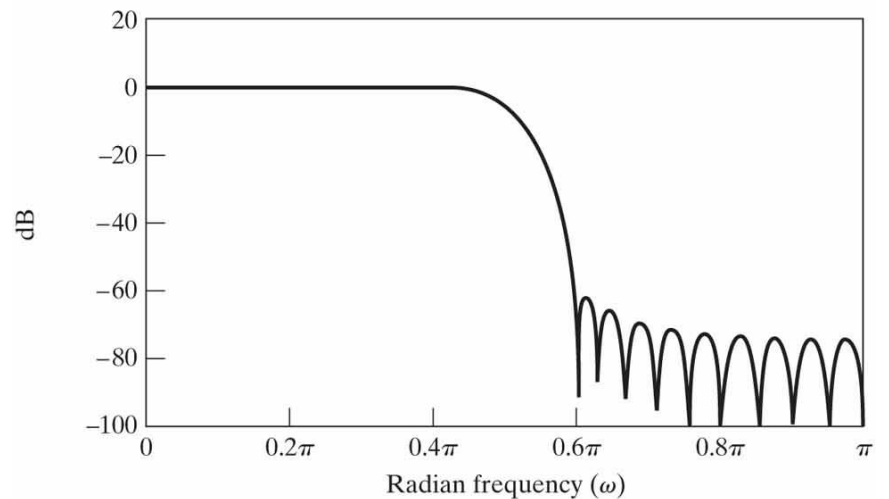
Amplitude



(a)



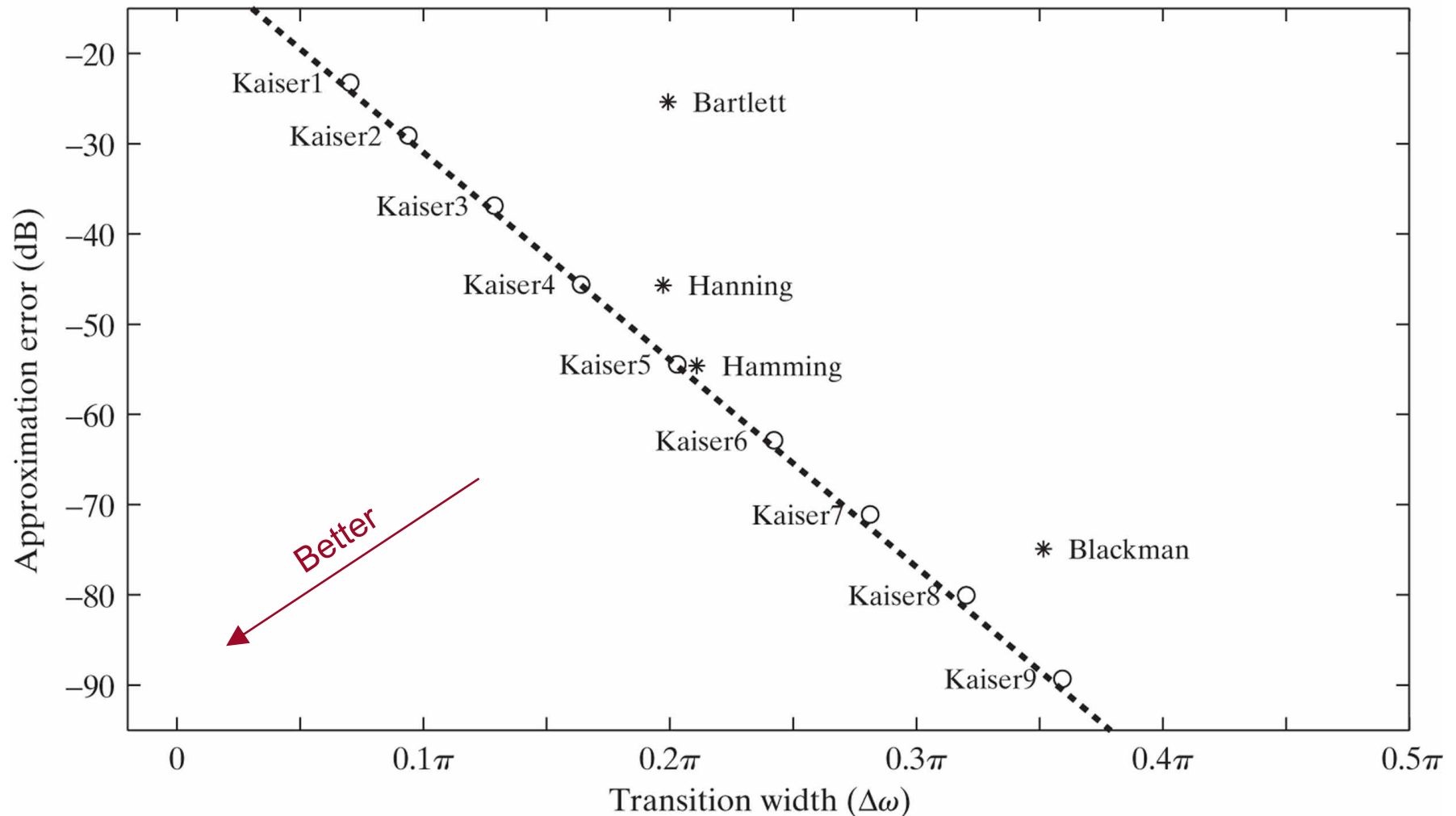
(c)



(b)

DT FIR FILTER DESIGN BY WINDOWING

Approximation error vs. Transition width [* = fixed windows, o = Kaiser (β = integer)]



LINEAR PHASE IN DT FIR FILTERS

- Although FIR filters usually require more complexity than IIR filters designed for the same specs, their advantage is the ability to provide (generalized) linear phase
- From the properties of DTFT, if a signal has the symmetry of the form

$$x[n] = x[M - n]$$

where M is the signal length, then

$$X(e^{j\omega}) = X_e(e^{j\omega}) \cdot e^{-j\omega M/2}$$

where $X_e(e^{j\omega})$ is a real (zero phase) even function.

- Hence, the phase is linear: $-\omega M/2$
- All windows $w[n]$ used in the window-based FIR filter design have this symmetry, so

$$W(e^{j\omega}) = W_e(e^{j\omega}) \cdot e^{-j\omega M/2}$$

LINEAR PHASE IN DT FIR FILTERS

- If the desired impulse response also has such symmetry

$$h_d[n] = h_d[M - n]$$

then

$$H_d(e^{j\omega}) = H_e(e^{j\omega}) \cdot e^{-j\omega M/2}$$

- The final filter's frequency response is ([see notes](#))

$$\begin{aligned} H(e^{j\omega}) &= H_d(e^{j\omega}) \odot W(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) W(e^{j(\omega-\theta)}) d\theta \\ &= \dots = A_e(e^{j\omega}) \cdot e^{-j\omega M/2} \end{aligned}$$

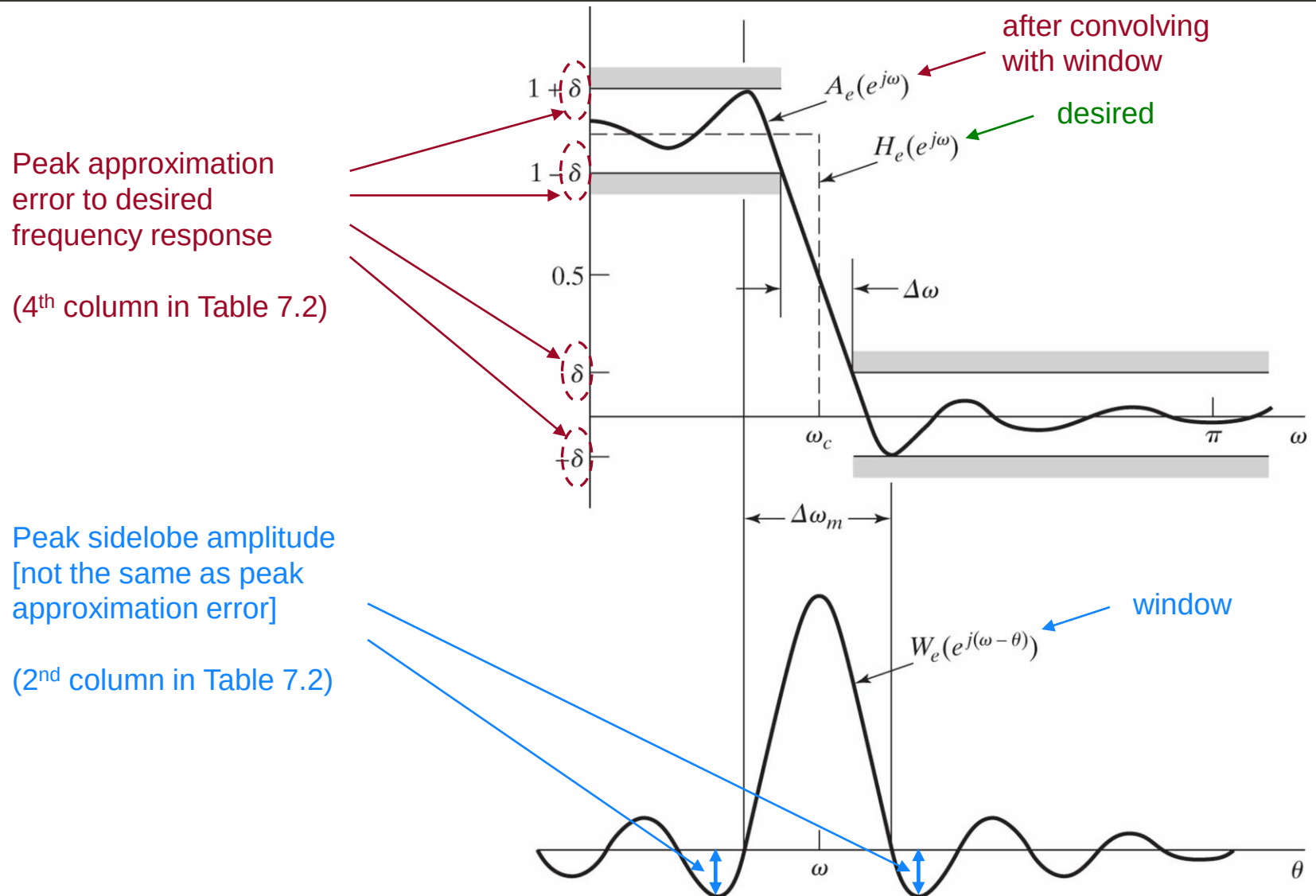
- Hence, the final FIR filter has a linear phase

DT FIR FILTER DESIGN BY WINDOWING

TABLE 7.2 COMPARISON OF COMMONLY USED WINDOWS

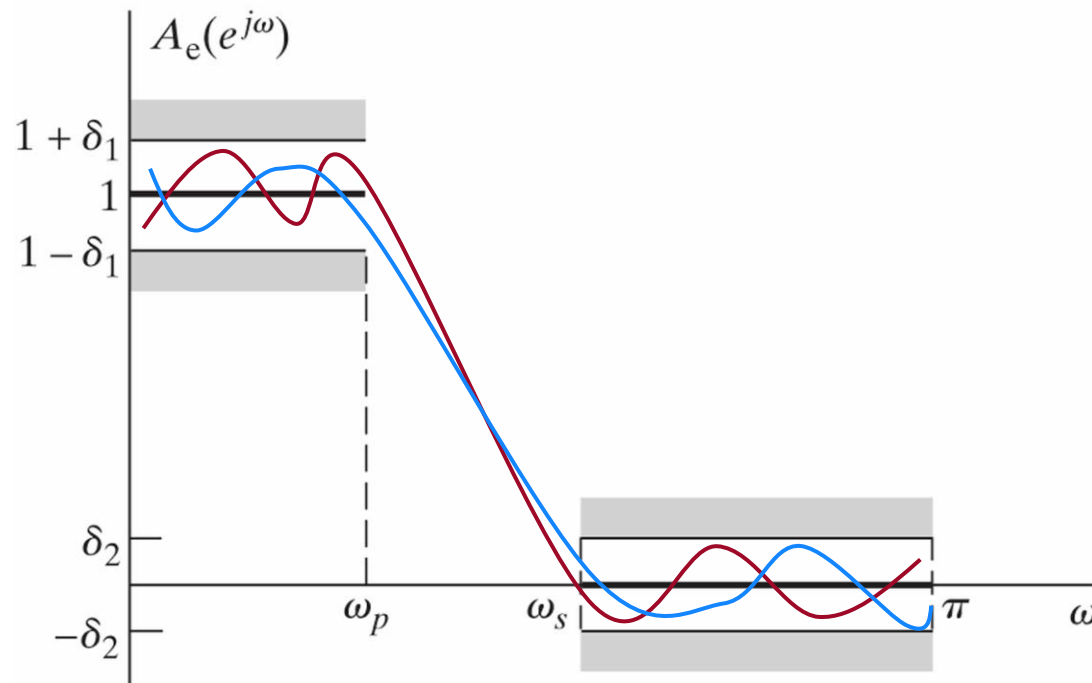
Type of Window	Peak Side-Lobe Amplitude (Relative)	Approximate Width of Main Lobe	Peak Approximation Error, $20 \log_{10} \delta$ (dB)	Equivalent Kaiser Window, β	Transition Width of Equivalent Kaiser Window
Rectangular	-13	$4\pi / (M + 1)$	-21	0	$1.81\pi / M$
Bartlett	-25	$8\pi / M$	-25	1.33	$2.37\pi / M$
Hann	-31	$8\pi / M$	-44	3.86	$5.01\pi / M$
Hamming	-41	$8\pi / M$	-53	4.86	$6.27\pi / M$
Blackman	-57	$12\pi / M$	-74	7.04	$9.19\pi / M$

DT FIR FILTER DESIGN BY WINDOWING



DT FIR FILTER DESIGN BY OPTIMIZATION

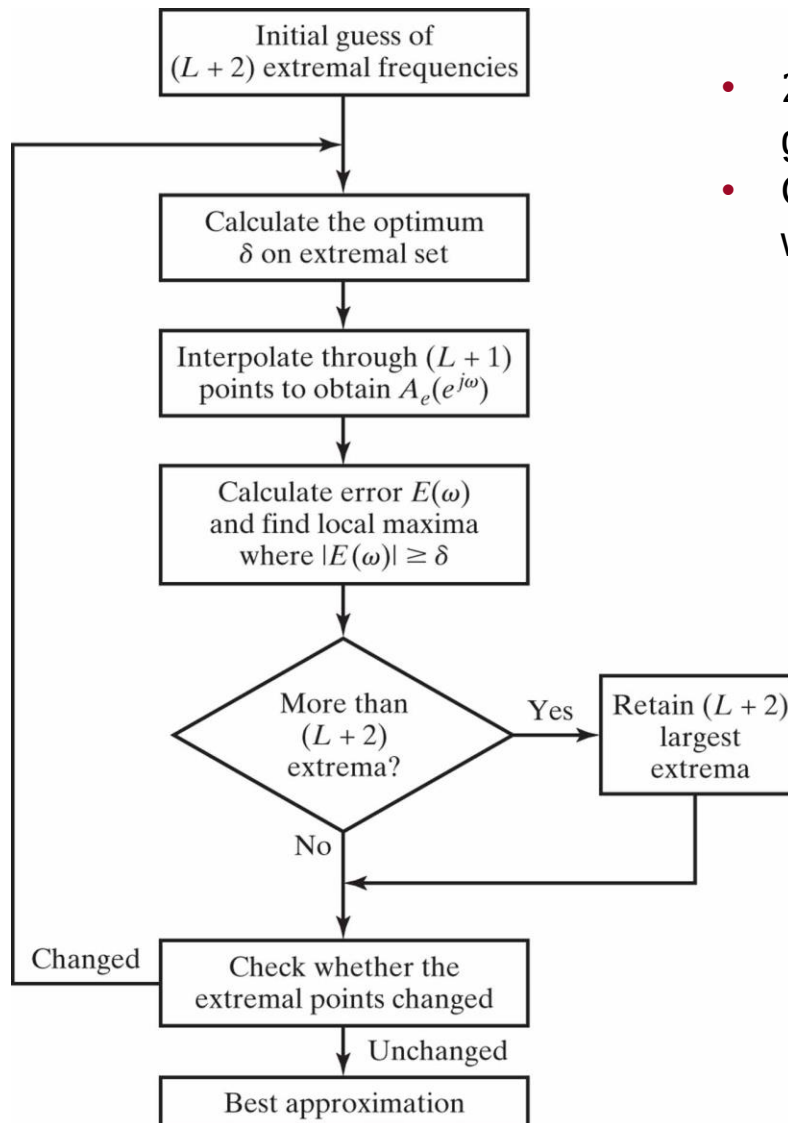
- FIR filters can also be designed by directly minimizing the deviation from the desired response $H_d(e^{j\omega})$
- For example, for a linear-phase lowpass filter, $H(e^{j\omega}) = A_e(e^{j\omega}) \cdot e^{-j\omega M/2}$, where $A_e(e^{j\omega})$ should approximate 1 in the passband and 0 in the stopband



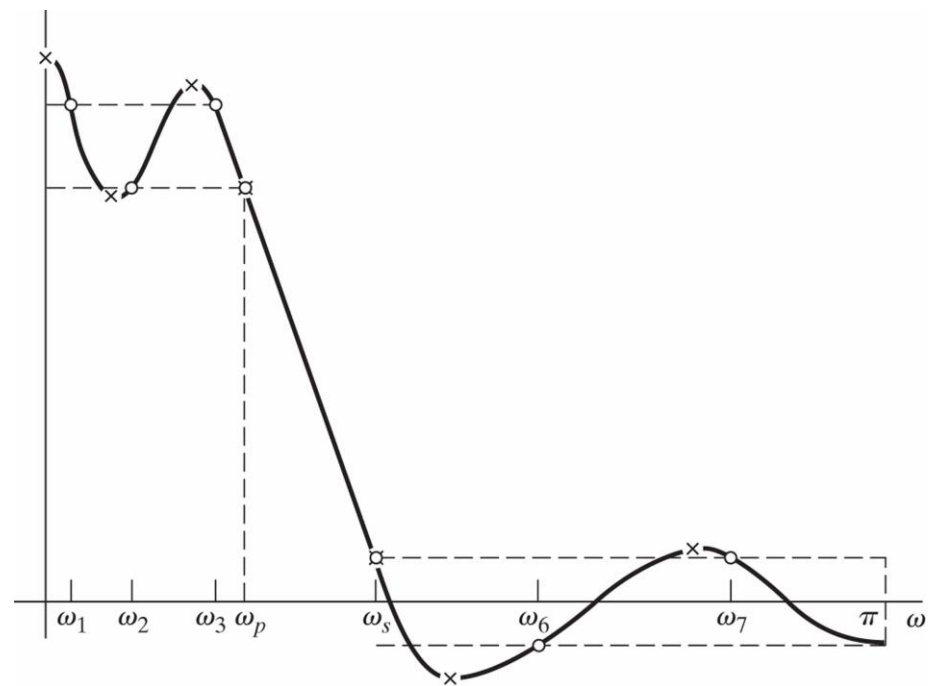
DT FIR FILTER DESIGN BY OPTIMIZATION

- A number of methods have been developed to find impulse response $h[n]$ that meets the specs while minimizing the deviation of $H(e^{j\omega})$ from $H_d(e^{j\omega})$
- Among these, the Parks-McClellan algorithm has become the most popular
- It utilizes the so-called *alternation theorem*, which relates minimax optimization to trigonometric polynomials (see textbook)
- The Parks-McClellan algorithm samples the frequency axis, fits a trigonometric polynomial to assumed approximation error peaks, then finds the actual error peaks and moves the frequency samples there, then repeats the iteration
- Details are in the textbook

PARKS-MCCLELLAN ALGORITHM



- $2L + 1$ is the length of the filter and needs to be guessed
- Could use $L = M/2$, where M is from Kaiser's window, as the initial guess



READING ASSIGNMENT

- Read Sections 7.1 – 7.5 in the textbook
- Start reading Chapter 8